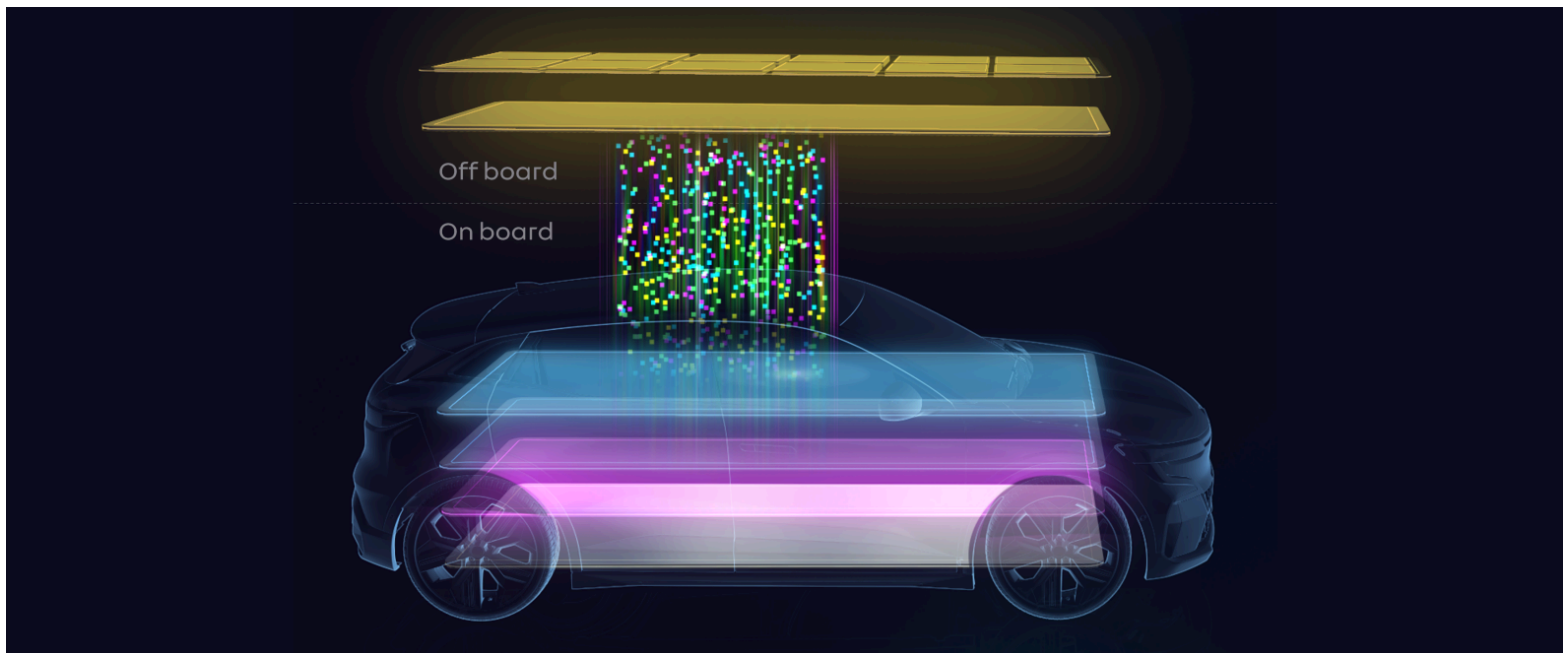


INNOVATION

All about Software Defined Vehicle

24 April 2023 ⌚ 3 MIN



PARTNERSHIP

SOFTWARE DEFINED VEHICLE

The Software Defined Vehicle (SDV) refers to the ability to upgrade a car throughout its lifetime, through a centralized architecture, updating it and integrating new applications to enhance its functions. This is the next revolution to which many car manufacturers like



Renault Group are preparing. These future cars defined around the software will work a bit on the same principle as a smartphone, but in a much more complex way! This new technology will equip the group's vehicles in the coming years. Explanations.

BY RENAULT GROUP



The Software Defined Vehicle, called SDV, represents the future of the automotive industry.

LUCA DE MEO
CEO Renault Group

[READ HIS LINKEDIN'S POST](#)

What is Software Defined Vehicle?

The Software Defined Vehicle (SDV) is the technological key that opens the field to give vehicles the possibility to evolve during their life: enhanced navigation and connectivity, upgrade applications in real time, enhanced safety and sustainability, etc. All for an improved driving experience. As cars are already becoming increasingly intelligent and connected, software will play an increasingly important role in both their design and evolution.



A continuous update of the car

Today, it is already possible to make remote updates of some vehicles via the **Firmware Over The Air (FOTA) system**. This is the case with the EASY-LINK multimedia systems of Clio, ZOE, Captur and Arkana and OpenR Link of **Mégane E-Tech electric and Austral**. This keeps the vehicle safe by making it easier and faster to improve the on-board system and apply patches.

Tomorrow, the **Software Defined Vehicle's flexible and scalable architecture** will enable the **faster development and integration of new features throughout the vehicle lifecycle, directly into the cloud**, that is, in secure online servers accessible from anywhere and anytime.

And the possibilities are almost endless. These new services can cover **preventive maintenance** of the car (with real-time detection of faults) as well as **personalization** of the interior of the passenger compartment, management of the battery charge or the **infotainment system**. In the end, it is up to each user to decide which functions they want to enjoy.





*Mégane E-Tech electric already has the OpenR Link system
updated with FOTA technology.*

What are the benefits of the Software Defined Vehicle?

Access to new features in real time

The Software Defined Vehicle will continuously **“upgrade” the vehicle by adding new features throughout its life cycle, in real time**. The offers that will be proposed may be collective or differentiated, in line with the real customs and driving habits of each. This will improve the user experience.

Enhanced safety on board

Thanks to the massive uptake of data collected from the car, the power of calculations and artificial intelligence offered by the Software Defined Vehicle, safety on board will be enhanced. For example, the wear and tear of certain parts or faults can be identified in real time, which will help to anticipate repairs or directly correct failures. We are talking about **predictive maintenance**.



An improvement in the value of the car

Being able to remotely upgrade the car via the cloud with the Software Defined Vehicle will bring an advantage in terms of driving experience and life on board, but also in terms of the value of the vehicle itself. Concretely, this means that after 3 or 4 years, the vehicle will have lost less value since it will have been enriched with new functions. Here we are talking about an **improvement in the residual value** of the vehicle.

Renault Group & Software Defined Vehicle techno...



With the Software Defined Vehicle, the electric and electronic architecture of the cars will be centralized and

How does Software Defined Vehicle works?

A central computer with much more power

Vehicles use an increasing number of on-board software and, to make them work, many electronic computers (processors). Today, a connected car has **between 60 and 80 computers**, all dedicated to a single function, such as vehicle lighting or driving aids. This represents **between 50 and 80 million lines of software code** in the various ECUs!

The current architecture of cars, with this abundance of computers spread over different parts of the vehicle, has become very complex. Problem: it is limited in terms of power and capacity to evolve the features of the car and integrate new ones.



With the Software Defined Vehicle, the idea is to have a **central computer with much more power, more robust and more flexible** than the car needs at its «birth». This will allow **to process the very large volumes of data collected by the various sensors of the car**, in particular from the driver assistance systems (ADAS), the body, the management of the chassis or the multimedia services and connectivity. But also to welcome the new functions over time without slowing down the system. To integrate this centralised super calculator capable of lasting over time, **Renault Group has formed a partnership with Qualcomm** and its «Snapdragon Digital Chassis» solution.

An operating system dedicated to the car

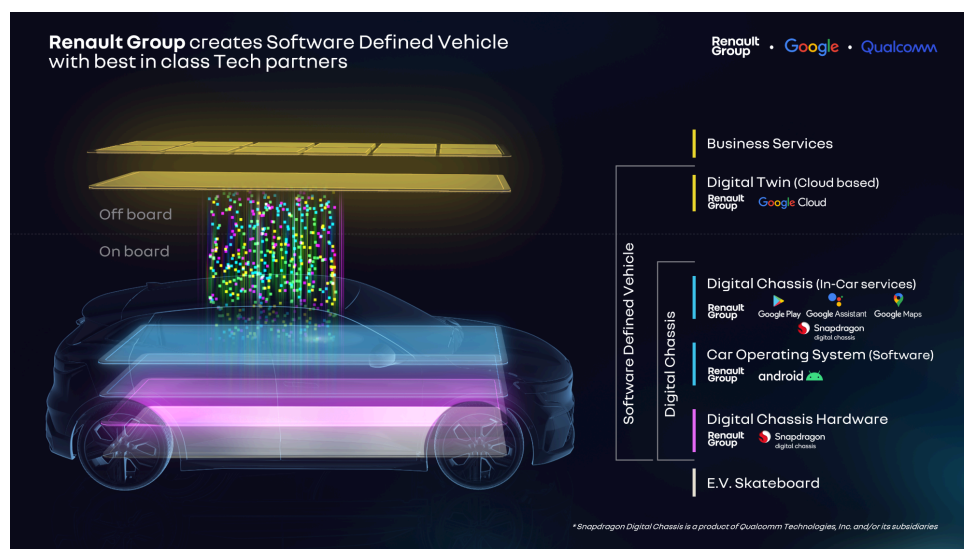
To add new features to the car, the Software Defined Vehicle relies on an operating system, a bit like in a computer or smartphone but adapted to the dashboard. It's called "CAR OS", for Car Operating System.

The new applications that will be developed will then be available in an application store and will be compatible with the car's operating system. To co-build this CAR OS, **Renault Group has chosen to work with Google**, a key partner in the Software Defined Vehicle project.



Secure data collection

The novelty of the Software Defined Vehicle is the increased amount of data related to the use of the car. This information makes it possible to identify and analyse the operation of the different areas of the car but also the behaviour of the users. Of course, the management of users' personal data is taken into account as well as all the necessary cybersecurity aspects.



Renault Group has chosen to co-develop the Software Defined Vehicle with state-of-the-art partners.

The Software Defined Vehicle is not just a vision, it is a reality and Renault Group has already entered the game.

To be continued...

Read also



INNOVATION

Vehicle Digital Twin: when physical and digital models unite

21 June 2022



EVENT

OpenR screen and OpenR Link – harnessing the latest technology in the All- new Renault Austral

08 March 2022

To keep up with Renault Group in real time



About us

[History](#)

[Brands](#)

[All our websites](#)

Events Websites

[Alpine Cars Events](#)

[BWT Alpine F1 Team Events](#)

[Dacia Events](#)

[Mobilize Events](#)

[Renault Events](#)

[Renault Group Events](#)

Regulations

[Reach regulation](#)

[Real Drive Emissions cycles](#)

Talent

[Our international vacancies](#)

[Our career areas](#)

[Grow in Renault Group](#)

Need help?

[Ask a question](#)

Other corporate websites

[Mobilize](#)

[Ampère](#)

© Renault Group 2024

[Cookies](#)

[Personal data](#)

[Legal infomation](#)

[Accessibility](#)

[Credits](#)

[Sitemap](#)

